

$$(7.2) \quad V \quad \phi(p) = 2p' + x^2 p''$$

$$p_0 = 2x^2 + 6x + C$$

$$p'_0 = 2 \cdot 2x + 6$$

$$\phi(p_0) = 4 \cdot 2x + 2 \cdot 6 + 2 \cdot 2x^2$$

$$p''_0 = 2 \cdot 2$$

V has dimension 3

$$\text{basis } B' = (1, x, x^2) \quad [\rho_0]_{B'} = \begin{pmatrix} C \\ 6 \\ 2 \end{pmatrix}$$

$$[\phi(p_0)]_{B'} = \begin{pmatrix} 2C \\ 4 \cdot 2 \\ 2 \cdot 2 \end{pmatrix}$$

$$\phi(1) = 0$$

$$\phi(x) = 2$$

$$\phi(x^2) = 4x + 2x^2$$

$$[\phi]_{B'}^{B'} = \begin{pmatrix} 0 & 2 & 0 \\ 0 & 0 & 4 \\ 0 & 0 & 2 \end{pmatrix}$$

$$\begin{vmatrix} -\lambda & 2 & 0 \\ 0 & -\lambda & 4 \\ 0 & 0 & 2-\lambda \end{vmatrix} = 2\lambda^2 - \lambda^3 = \lambda^2(2-\lambda)$$

$$\lambda_1 = 0 \quad \text{m.s. 2}$$

$$\lambda_2 = 2 \quad \text{m.s. 1}$$

$$\rightarrow (\text{Im } A \cap \text{Ker } A) \neq \emptyset$$

$$\begin{array}{l} 0 \\ 2 \end{array} \quad \begin{pmatrix} 0 & 2 & 0 \\ 0 & 0 & 4 \\ 0 & 0 & 2 \end{pmatrix} \xrightarrow{\text{d.i.}} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \xrightarrow{\phi \cdot \lambda \cdot id} \begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix} \mapsto \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} -2 & 2 & 0 \\ 0 & -2 & 4 \\ 0 & 0 & 0 \end{pmatrix} \xrightarrow{\text{d.i.}} \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix}$$

$$\begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix} \mapsto \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

$$\text{Indíř } B = \left(\begin{pmatrix} 2 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 \\ 2 \\ 1 \end{pmatrix} \right) = (2, x, x^2 + 2x + 2)$$

$$J = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 2 \end{pmatrix} = [\phi]_B^B$$

Komentář: využití izomorfismu mezi v.p. V a $\mathbb{R}^3$, pomocí
drivní najdu matici zobrazení a tu vyřeším: klesíky viz cvičení.

Na konci přivedu zpět do V .